

Customer Shopping Behavior Analysis

Transactional insights from **3,900 purchases** across product categories — spending patterns, customer segments, and subscription behavior.



Dataset at a Glance

3,900

Purchases

18

Columns

37

Missing Values

Review Rating

\$59.76

Avg. Purchase

Demographics · Purchase details · Shopping behavior

Data Preparation in Python

01

Load & Explore

pandas, df.info(), .describe()

02

Clean

Impute missing Review Ratings by category median

03

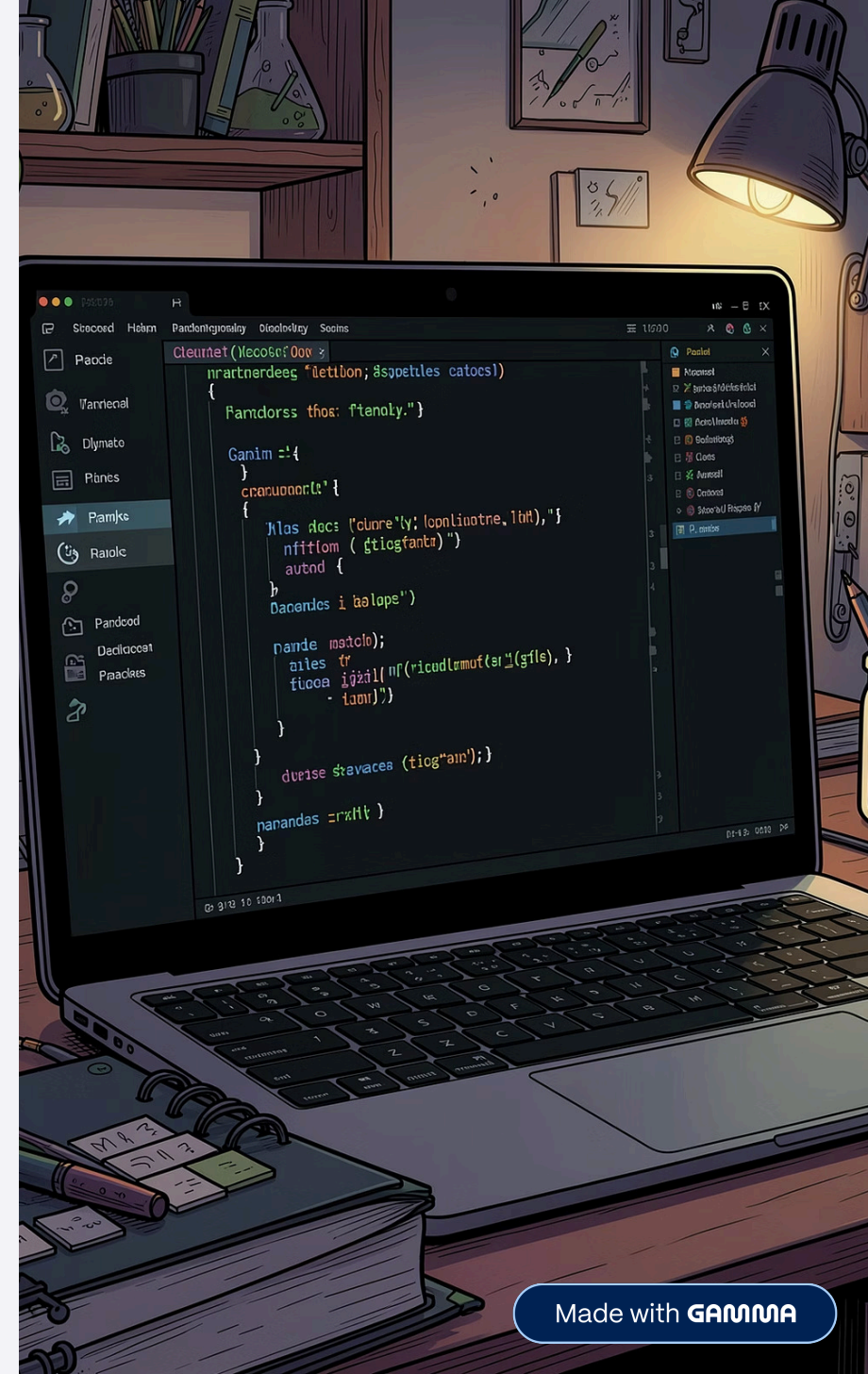
Engineer

Created age_group and purchase_frequency_days

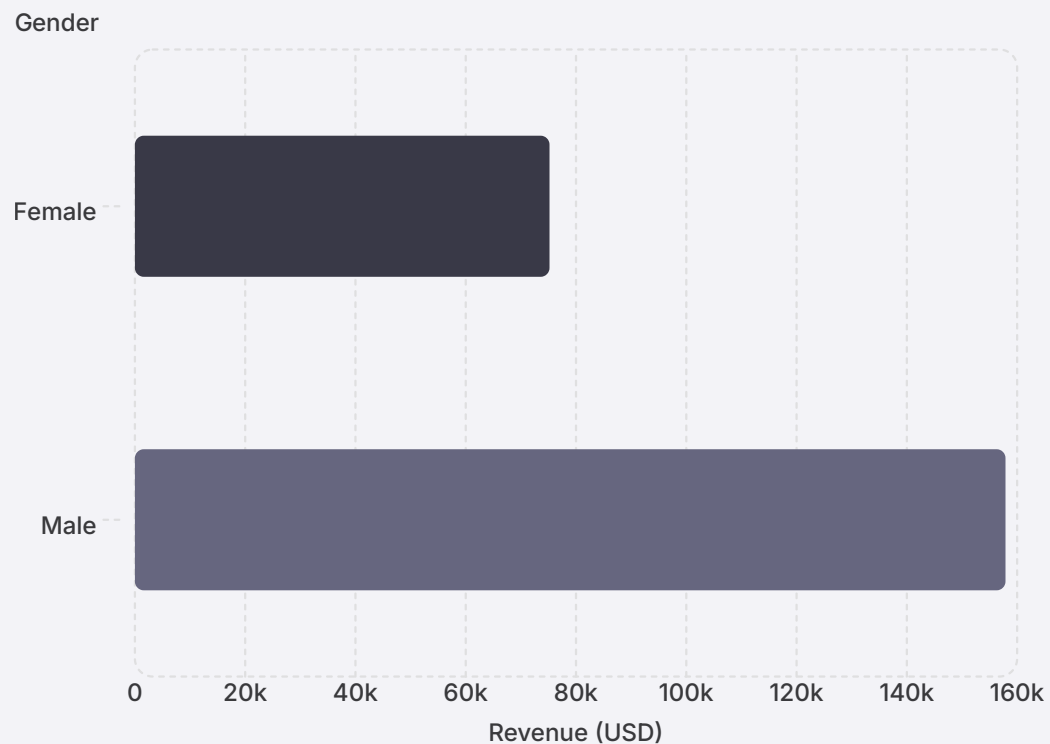
04

Load to DB

Connected to PostgreSQL for SQL analysis



Revenue by Gender



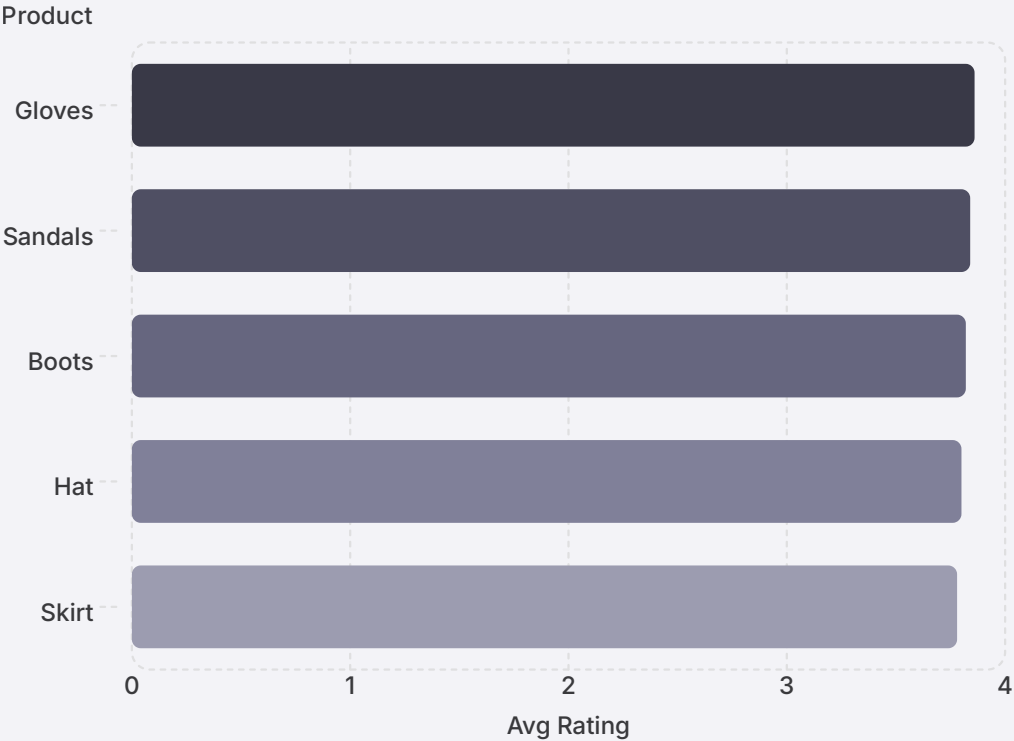
Male customers drive 2× revenue

Male: **\$157,890** vs. Female: **\$75,191**

Male shoppers also represent **68%** of all customers (2,652 vs. female share).

Key Product & Shipping Insights

Top 5 Products by Rating



Shipping Type Comparison

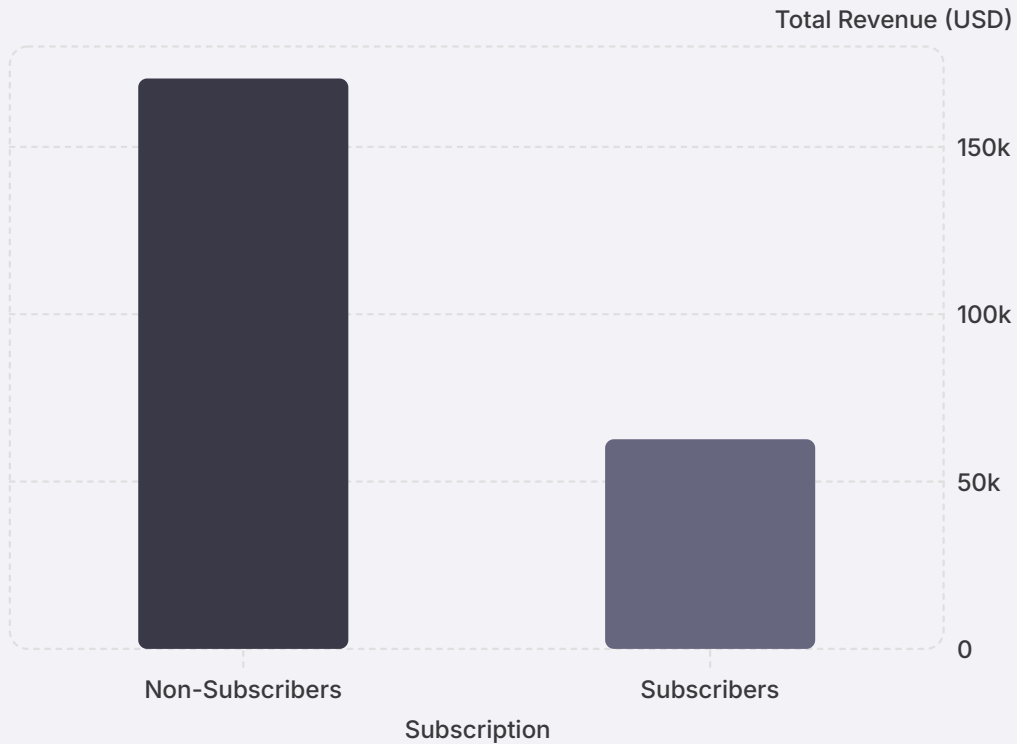
Standard: **\$58.46** avg

Express: **\$60.48** avg

Most Discount-Dependent

Hat (50%), Sneakers (49.7%), Coat (49.1%)

Subscribers vs. Non-Subscribers



73% are non-subscribers

Yet average spend is nearly identical: **\$59.87** vs. **\$59.49**

Among repeat buyers (5+ purchases): **958** subscribe vs. **2,518** don't.

i Subscribers are a high-potential growth segment.



Customer Segmentation

Loyal

3,116 customers

Returning

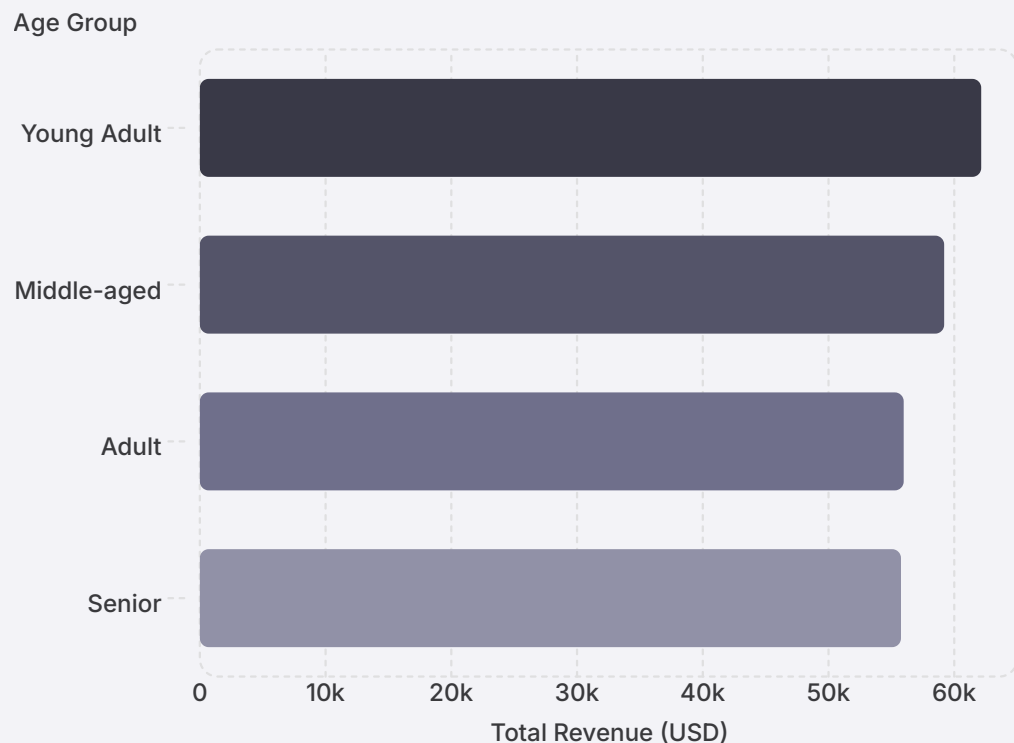
701 customers

New

83 customers

80% of customers are already in the Loyal segment — a strong retention foundation.

Revenue by Age Group



Young Adults lead revenue

Young Adults contribute **\$62,143** — the highest among all age groups.

Revenue is relatively evenly distributed across the four groups, with only a **\$6,380** gap between highest and lowest.

Power BI Dashboard



Interactive Dashboard Highlights

- **3.9K** customers · **\$59.76** avg purchase · **3.75** avg rating
- Filters: Subscription, Gender, Category, Shipping
- Charts: Revenue & Sales by Category and Age Group
- Subscription split: 27% Yes · 73% No



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward "Loyal" segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users